Umair Javaid Manj

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Research Experience

Doctoral Researcher — **Auckland University of Technology**, Health & Artificial Intelligence, Auckland — Oct 2025 to Present

- Developing an AI based biomarker of chronic low back pain from cortical responses to paraspinal stimulation.
- Adapted the Zurich paraspinal vibration fMRI paradigm (Cole et al., 2022; Dörig et al., 2026) to EEG, comparing painful and healthy spinal segments within the same person.
- Designing a case control EEG study combining source imaging, representational similarity analysis and subject level machine learning.
- Led the first pilot and identified an analysis artefact behind an apparent decoding effect.

Data & AI Governance Analyst — **New Zealand College of Chiropractic**, Auckland — Aug 2026 to Present

- Analyse EEG hyperscanning of chiropractor and patient: inter brain and within brain connectivity (wPLI, dPLI) and source level network analysis.
- Apply graph theory to resting state EEG networks in a randomised chiropractic trial of adults and children, with linear mixed models.
- Model longitudinal clinical outcomes (immune response, DASS-21, PROMIS-25) with mixed effects models in R.
- Manage private participant data and consent records.

Instructor — **NUST**, Islamabad — May 2023 to July 2024

- Supervised 18 student term projects; awarded Best Instructor.

Research Associate — **COGNET Lab, SEECS**, NUST, Islamabad — Aug 2021 to Apr 2023

- Researched the robustness of deep learning and computer vision models to adversarial inputs.
- Mentored junior research assistants (Research Assistant until Nov 2022).

Publications

- Akhter, M. N., Ali, H., Niazi, I. K., **Manj, U. J.**, & Taylor, D. (2026). Application of machine learning techniques in the measurement of ground reaction force (GRF) waveforms using inertial measurement unit (IMU) sensors: a systematic literature review. *Medical Engineering & Physics*. doi
- Rizvi, D. Q., Majeed, M., **Manj, U. J.**, & Haq, M. I. U. (2025). Sleep disorder prediction with PSO-optimized machine and deep learning models. *ICODT2*, pp. 1 to 6. doi
- Akhter, M. N., **Manj, U. J.**, Ali, H., Niazi, I. K., & Taylor, D. Deep learning-based kinetics and center of pressure estimation during sit-to-walk transitions from a single sacrum-mounted IMU. *IEEE Transactions on Biomedical Engineering* (under review).
- Akhter, M. N., Ali, H., Niazi, I. K., **Manj, U. J.**, & Taylor, D. Predicting ground reaction force during sit to walk: towards kinetically instrumenting the Timed Up and Go test. *Medical & Biological Engineering & Computing* (under review).
- **Manj, U. J.**, Rizvi, S. D. Q., Niazi, I. K., Akhter, M. N., & Ali, H. Gated feature fusion and age confounding in plantar thermography for diabetes screening. *Biomedical Physics & Engineering Express* (under review).
- **Manj, U. J.**, Qamar, D., Rashid, M. A., Perwaiz, N., & Niazi, I. K. Speaker leakage and subject level accuracy in voice biomarkers for Parkinson's disease. *Machine Learning: Health* (under review).

Education

PhD	**Auckland University of Technology**, Health & Artificial Intelligence, Auckland, New Zealand	Oct 2025 to Present
MS	**National University of Sciences & Technology (NUST)**, Computational Science & Engineering, Islamabad • Thesis: energy optimisation in data centres using fuzzy inference systems (~30% cost reduction).	2017 to 2021
BE	**National University of Sciences & Technology (NUST)**, Computer Engineering, Islamabad • Thesis: NATO narrowband waveform on a software defined radio (digital signal processing, convolutional encoding).	2012 to 2016

Skills

Neurophysiology & Signals: EEG and IMU analysis, source localisation, functional connectivity and graph theory, hyperscanning, vibrotactile stimulation

Machine Learning & Statistics: subject level evaluation, leakage and confounding checks, SHAP, linear mixed models, cluster permutation tests, G*Power

Programming: Python (MNE, PyTorch, scikit-learn), R, MATLAB (EEGLAB), C++

Research Practice: Participant data governance, systematic review, scientific writing

Referees

Dr. Nazia Perwaiz, MSCA Industrial Postdoctoral Fellow, School of Computer Science, University of Galway, Ireland; Assistant Professor, SEECS, NUST. nazia.perwaiz@seecs.edu.pk

Dr. Haider Ali, Faculty of Design and Creative Technologies, Auckland University of Technology; AI Engineer, KPMG New Zealand. haider.ali@autuni.ac.nz